

- 5 (a) State what is meant by simple harmonic motion.

.....

.....

..... [2]

- (b) A block is suspended by a spring. The block oscillates vertically with simple harmonic motion.

The velocity v of the block varies with time t according to

$$v = 0.56 \cos 16t$$

where v is in m s^{-1} and t is in s.

- (i) Calculate the period of the oscillation.

period = s [1]

- (ii) Determine the amplitude x_0 of the oscillation.

$x_0 = \dots\dots\dots$ m [2]

- (iii) Use your answer in (b)(ii) to determine the equation for v in terms of the displacement x of the block, where v is in m s^{-1} and x is in m.

$v = \dots\dots\dots$ [1]

